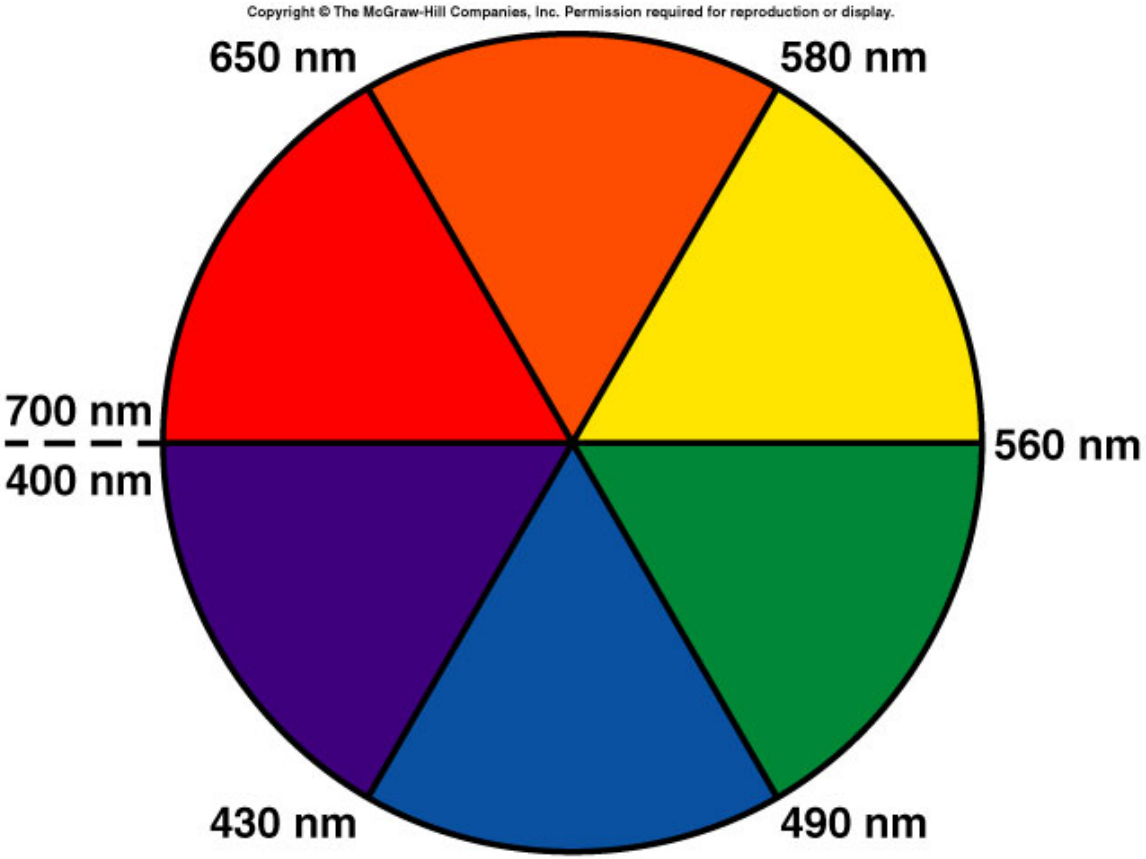
화학특강 (2009년 2학기)
제5장

황현석

2009년 10월 12일

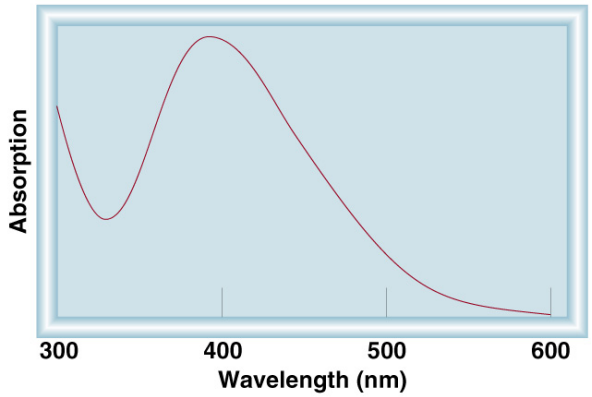


Complementary Colors



Absorption and Fluorescence

- Absorption spectrum of bromine



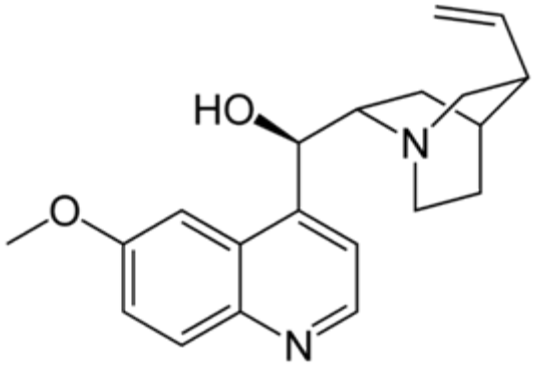


$Br_2(aq) + HCOOH(aq) \rightarrow 2Br^- + 2H^+(aq) + CO_2(g)$

Chang, 13.2

- Fluorescence of quinine

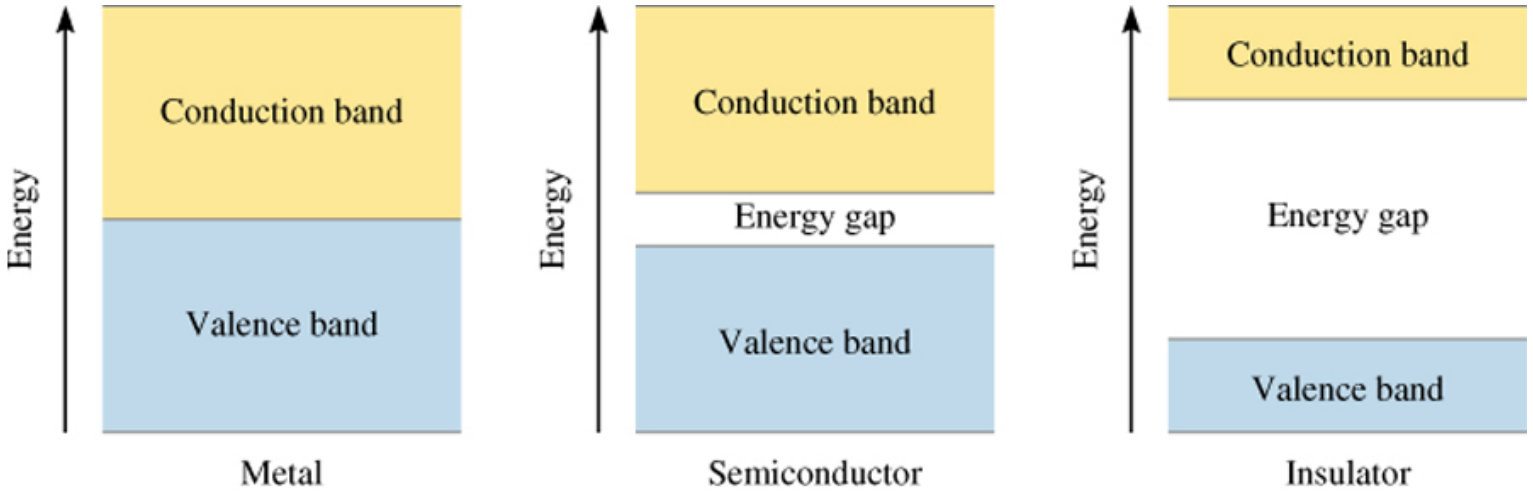




quinine

Semiconductor

- 반도체 (semiconductor): 전도체 (conductor)와 부도체 (insulator) 사이에 존재하는 물질로서, 일정한 조건에서 도체가 되는 물질.



Quantum Dots (QD) – an Artifical Atom

- Energy of a particle in a box $$E_n = \frac{n^2 h^2}{8 m_e L^2}$$

- Fluorescence of quantum dots

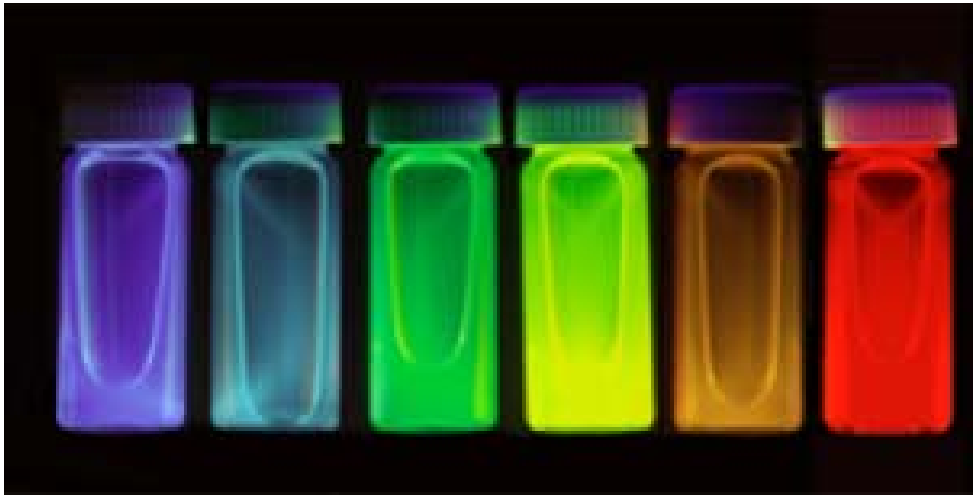
$$E_{gap}^{QD} = E_{gap}^{Bulk} + \frac{\pi^2 h^2}{2R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right) - \frac{1.8e^2}{\varepsilon R}$$

Quantum Dots (QD) – an Artifical Atom

- Energy of a particle in a box

$$E_n = \frac{n^2 h^2}{8 m_e L^2}$$

- Fluorescence of quantum dots



Colloidal CdSe quantum dots dispersed in hexane. Quantum confinement effects allow quantum-dot color to be tuned with particle size.

Quantum Dots (QD) – an Artifical Atom

- Interior of CdSe quantum dots

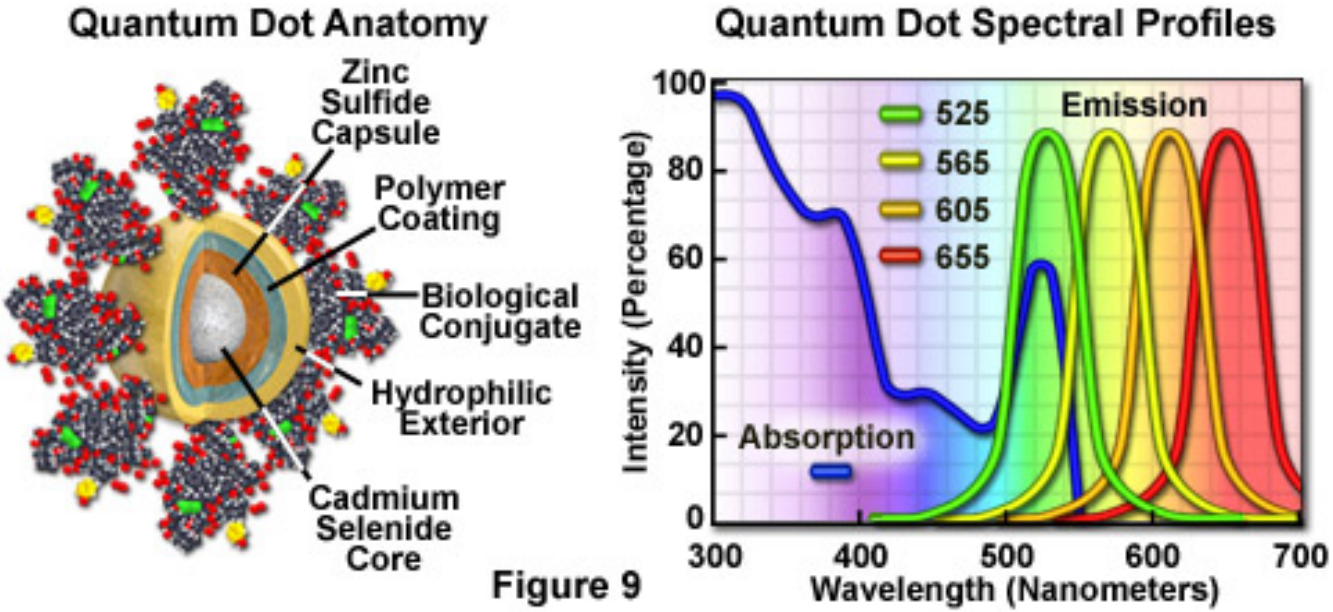


Figure 9

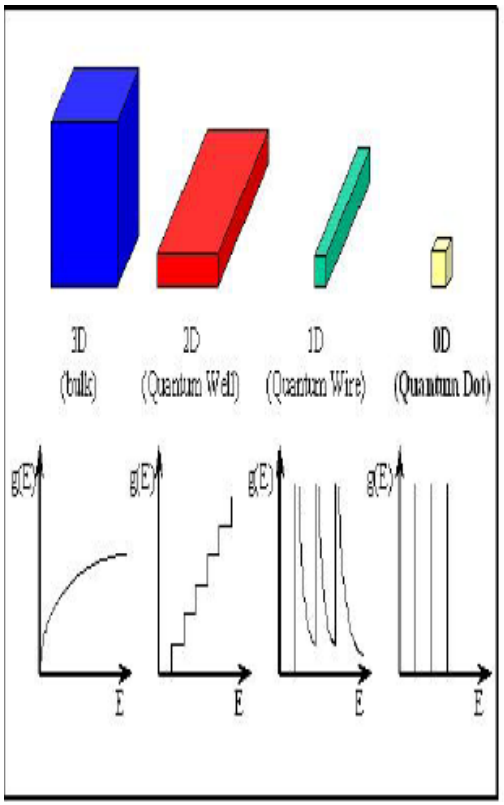
- Two disadvantages of CdSe QDs
  1. High toxicity – coating the QD's in ZnS to prevent the CdSe crystal from dissolving
  2. Insolubility in water – coating with a water-soluble hydrophilic polymer

Characteristics of Quantum Dots (QD)

- Quantum dots are small devices that contain a tiny droplet of free electrons.
- Typical dimensions are between nanometers to a few microns.
- A quantum dot can have anything from a single electron to a collection of several thousands.
- The size ,shape and number of electrons can be precisely controlled.
- Just as in an atom, the energy levels are quantized due to the confinement of electrons.
- The 3D spatial confinement is observed in the quantum dots.
- Unlike atoms however, quantum dots can be easily connected to electrodes and are therefore excellent tools to study atomic-like properties…..

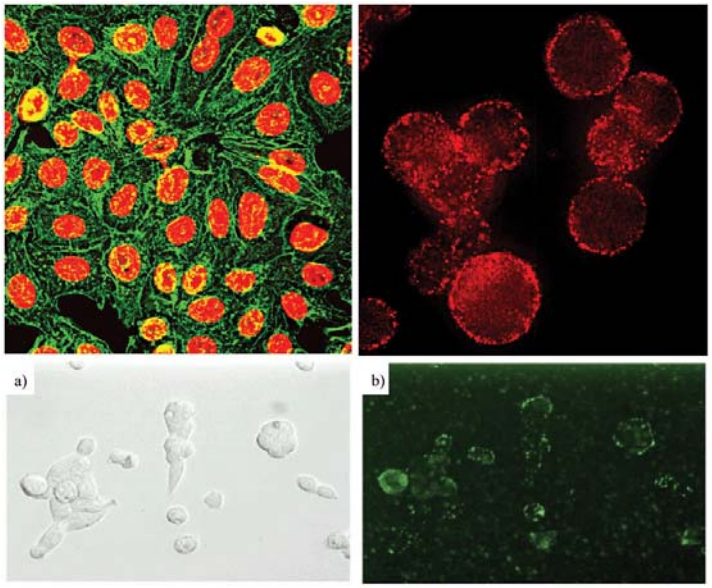
History of Quantum Dots

- In the 1970s the first low dimensional structures Quantum Wells were developed.
- 1D (quantum wires) and 0D (quantum dots) were subsequently developed.

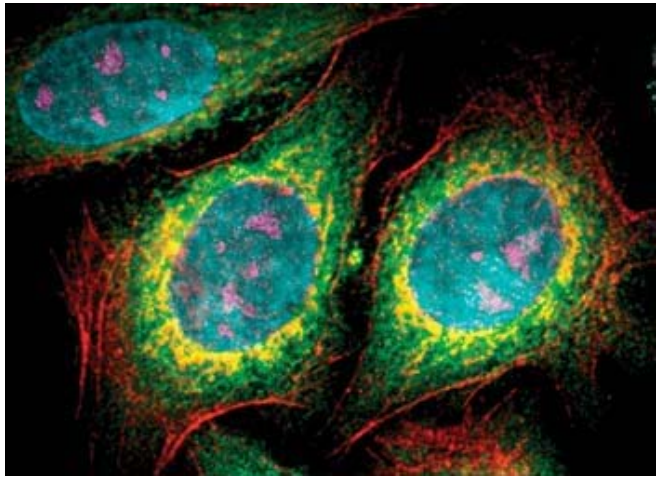


Quantum Dots and Their Applications (I)

1. Quantum dots in biology and medicine
   - Advantages of QD over organic dye molecules: organic dye molecules bleach within a minute or two, while QD's do not bleach and they are also about 10- to 20-times as bright as commonly used organic fluorophores.



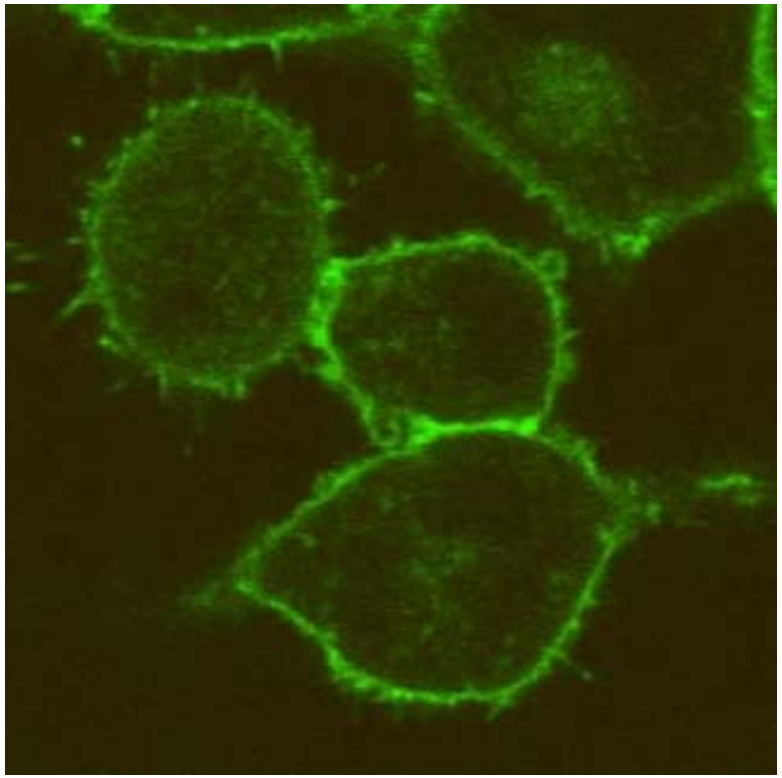
Parak et al., Nanotechnology **16**, R9 (2005)



the nucleus is blue, a specific protein within the nucleus is pink, mitochondria look yellow, microtubules are green, and actin filaments are red . (Quantum Dot Corp.)

Quantum Dots and Their Applications (II)

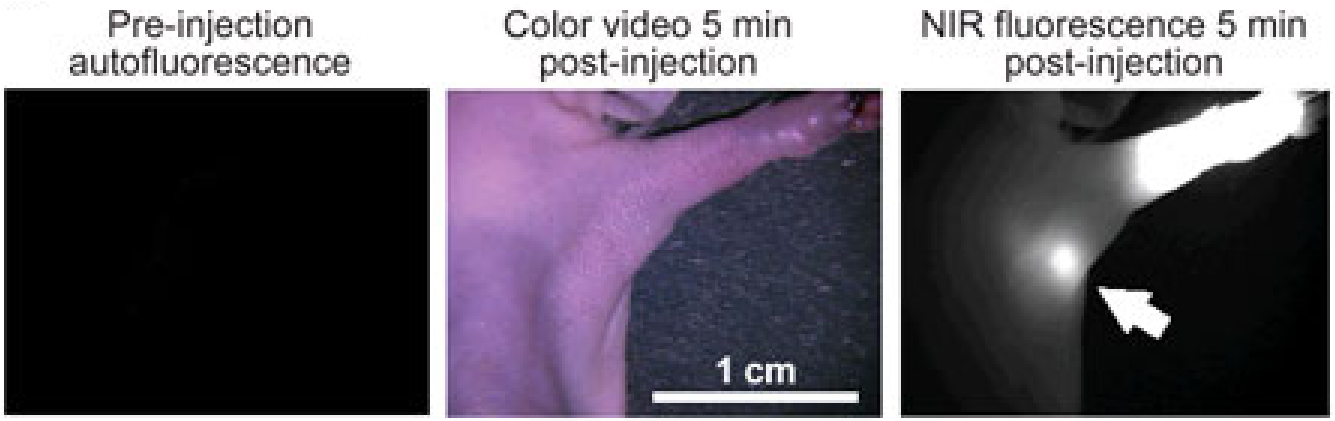
- Quantum dots to study living cells



Lidke et al., Nature Biotechnology **22**, 198 (2004)

Quantum Dots and Their Applications (III)

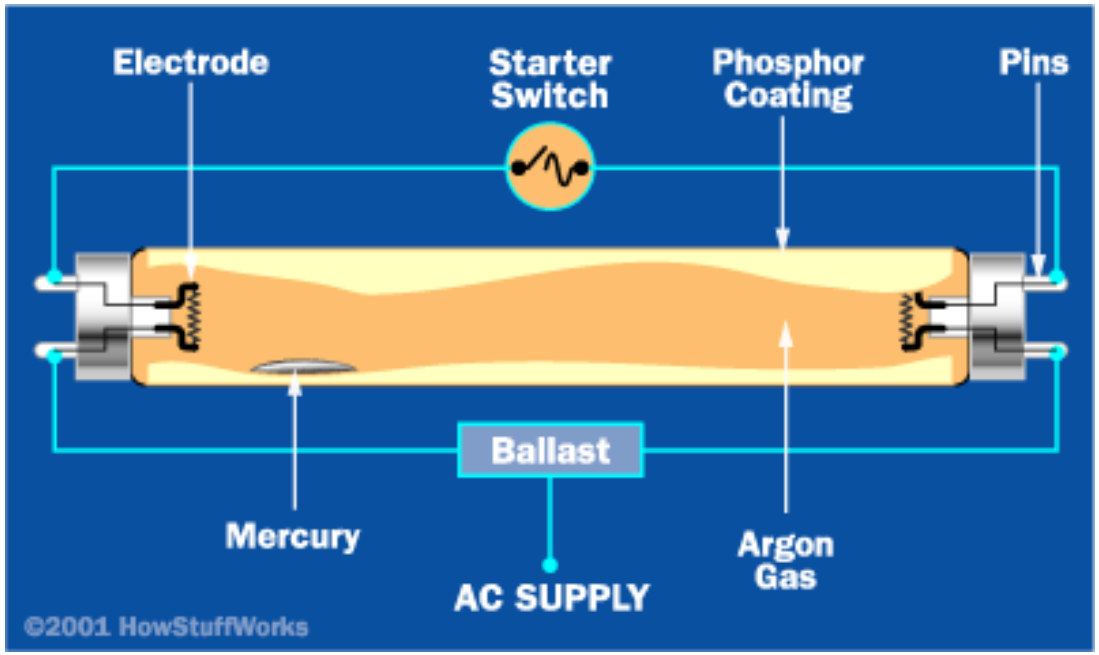
- A limitation of the fluorescent dots is that visible light frequencies would only be useful near the surface of the skin.
  → QD's that emit near-infrared light.
  → Near infrared penetrates tissue much more efficiently than visible light.



Kim et al., Nature Biotechnology **22**, 93 (2004)

Quantum Dots and Their Applications (IV)

2. Quantum dots in Electronics
    - Increasing the efficiency of fluorescent lighting.



Conventional fluorescent lamp

Quantum Dots and Their Applications (V)

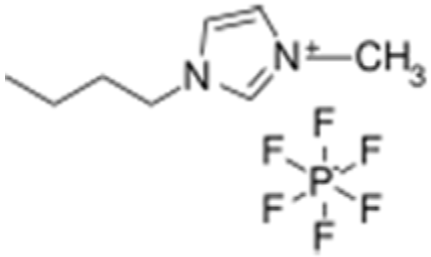
2. Quantum dots in Electronics
    - Increasing the efficiency of fluorescent lighting.



QD fluorescent lamp

Ionic Liquids

- (Room temperature) Ionic liquids: liquids that contains essentially only ions at normal temperature.
- Properties of ionic liquids
  1. Composed of small anions such as BF_4^- (tetrafluoroborate) or PF_6^- (hexafluorophosphate) and large cations such as1-hexyl-3-methylimidazolium or 1-butyl-3-methylimidazolium.



  2. Remain liquids because the bulky, asymmetrical cations do not pack together efficiently with the smaller, symmetrical anions.
  3. Are electrically conductive and have extremely low vapor pressure.

Ionic Liquids

- Advantages of ionic liquids
  1. Their long liquid ranges, typically from -100˚C ~ 200˚C.
  2. The environmentally friendly nature.
  3. Can replace organic solvents.
  4. Cations that will attract potentially harmful ions such as mercury, cadmium, uranium, and americium (the latter two found in nuclear waste materials).
  5. Cations that will remove H_2S (which produces SO_2 when gas is burned) and remove CO_2 from the exhaust gases of fossil-fuel-burning power plants.
- Disadvantages of ionic liquids
  1. Their cost – very expensive.

Ionic Liquids

